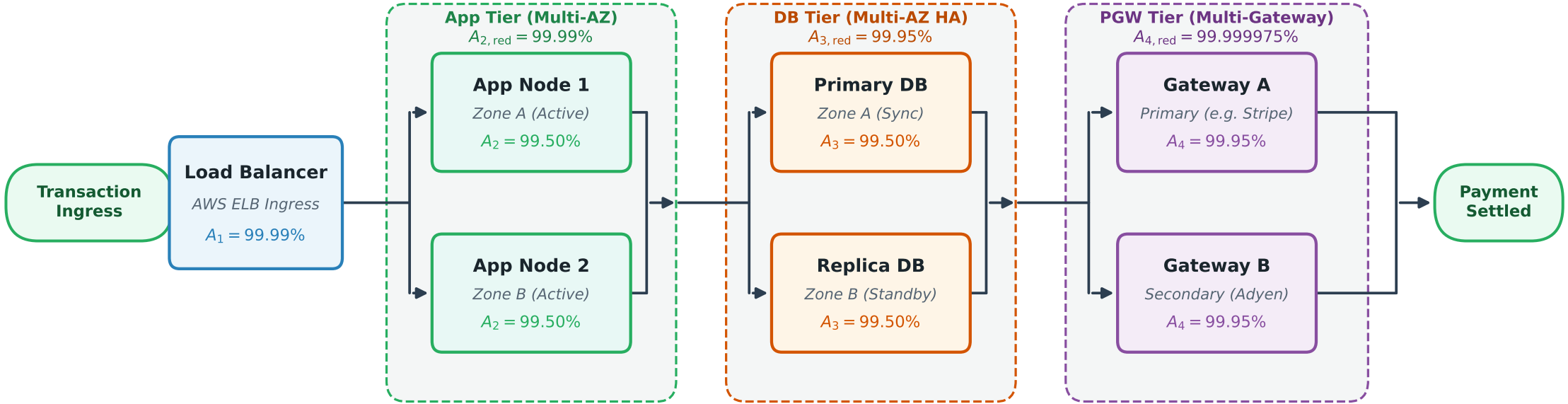
Scenario 3: Multi-Gateway Expansion Architecture (End-to-End Fault Tolerance)

$A_{\text{scen3}} = A_1 \cdot A_{2,\text{red}} \cdot A_{3,\text{red}} \cdot A_{4,\text{red}} = 99.930\% \implies$ 6.13 h/year downtime (−93.4% vs. baseline)



Architectural Peak: Dual external gateway routing reduces annual third-party downtime to 0.002 h/year (approx. 8 seconds/year).